

Pavan Yadava Annappa

Machine Learning Engineer - 3D Perception, Sensor Fusion & Edge Deployment

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3+ years in automotive ADAS and perception (Continental AG / AUMOVIO), currently researching end-to-end autonomy alongside an M.Sc. in AI for Autonomous Driving at FAU Erlangen-Nürnberg. I build perception systems the whole way down: BEV LiDAR-camera-radar fusion, transformer encoders and multitask heads, then ONNX, TensorRT FP16/INT8, ROS 2 C++, Rust and OTA fleet rollout, until it runs in real time on hardware I can measure. Every figure below was measured on the hardware named beside it.

SELECTED PROJECTS

LangGrasp github.com/pavanyadava007/langgrasp

"Pick the blue screwdriver" becomes a box, a mask, a grasp pose and a motion on a simulated SO-101 arm.

115/120 placed full language pipeline, 120 stratified trials **682 ms per command** median end to end **-9 placements**

cost of removing the HSV colour check

Grounding DINO · YOLO11n-seg · TensorRT · ONNX · MuJoCo · ROS 2 Humble · LeRobot ACT · PPO

offroad-bevfusion github.com/pavanyadava007/offroad-bevfusion

Camera + LiDAR + radar multitask BEV perception, exported to TensorRT and deployed as a ROS 2 C++ node.

36.4 → 15.2 / 14.7 ms PyTorch FP32 → TensorRT FP16 / INT8 **11.5 ms/frame** ROS 2 Humble C++ node on the FP16 engine

mAP 0.022 → 0.053 camera-only → cam+LiDAR+radar ablation

PyTorch · LSS · CenterPoint · deformable attention · ONNX opset 17 · TensorRT FP16/INT8 · ROS 2 Humble · C++

AeroEdge github.com/pavanyadava007/aeroedge

One aerial-detection model compressed to 3.2 MB INT8 and shipped to a 20-device edge fleet through signed OTA canary rollouts.

10.61 → 3.17 MB FP32 → INT8 QDQ ONNX (3.3x) **90.6 s / 4.7 s** canary 5% → 50% → 100% across 20 live agents, 0 failures

p95 29.6 ms · 34 FPS INT8 @640, ORT 8 threads

YOLO11n · ONNX Runtime · INT8 PTQ · MQTT · Docker · Rust (ort 2.0-rc.13) · FastAPI · GitHub Actions

Makerspace Reuse Scanner github.com/pavanyadava007/makerspace-reuse-scanner

An edge camera detects reusable parts on a belt and an on-premise German/English RAG assistant answers disposal questions with citations.

20/20 · 3/3 · 2/2 RAG eval, 25 hand-verified questions **2.44 s median** per question, warm llama3.1:8b, k=4 **mAP50 0.410**

detector on the held-out test split

YOLO11n · ONNX Runtime · FastAPI · Postgres/pgvector · Ollama llama3.1:8b · nomic-embed-text · React · Docker Compose

SIGINT-Fusion github.com/pavanyadava007/sigint-fusion

Simulated RF sensors stream I/Q through a Java ingest gateway and Kafka into an ONNX modulation classifier, an emitter fuser and a RAG analyst agent.

75.7% top-1 21 modulations, SNR -10 to 30 dB **87.6% ± 4.3** 5-shot on 3 modulations never seen in training

hit@5 100% · MRR 0.93 hybrid dense + full-text retrieval

Spring Boot 3 · Java 21 · Kafka · ONNX Runtime · ResNet-1D · LangGraph · MCP · Postgres/pgvector

EXPERIENCE

AI / ML Engineer and Researcher, Automotive Perception Oct 2025 - present

AUMOVIO SE (Continental AG automotive spin-off) · Frankfurt, Germany

- SparseDrive end-to-end model research: vulnerable-road-user detection and point-cloud pre-processing in C++/Python.
- MLOps evaluation tooling reporting mAP and precision-recall per test scenario.
- Radar test automation.

AI & Data Science Intern, Automotive Division Apr 2024 - Oct 2025

Continental AG · Alzenau, Germany

- ADAS validation workflows with deep learning, sensor fusion and image/video processing.
- Real-time visualisation tooling.
- Docker-containerised deployment.

Engineer, ADAS Software Algorithm Development

Jun 2022 - Sep 2024

Continental AG · Bengaluru, India

- CV and Transformer models for series Traffic Sign Recognition.
- Landmark detection, automatic camera calibration and ODD testing.
- KPI-evaluation pipelines saving 30+ hours/week, under ISO 26262 (ASIL-B) and MISRA C++.

EDUCATION

M.Sc. Artificial Intelligence in Autonomous Driving & Electromobility

Oct 2024 - present

Friedrich-Alexander-Universität Erlangen-Nürnberg · Erlangen-Nürnberg, Germany

B.E. Electronics and Communication Engineering

2021

The National Institute of Engineering · Mysuru, India

SKILLS

Perception & Signals	BEV perception (LSS, deformable attention) · 3D detection (CenterPoint, PointPillars) · Semantic segmentation · Occupancy prediction · Multitask learning · LiDAR-camera-radar fusion · End-to-end autonomy · 2D detection (YOLO11) · Domain-shift evaluation · RF modulation classification · Classical DSP (PSD, STFT, CFAR) · Kalman/Hungarian multi-target fusion · Few-shot and self-supervised (BYOL) transfer
Deployment & Compression	TensorRT (FP16 / INT8 PTQ) · Static-shape ONNX export and parity testing · ONNX Runtime (Python, Rust, WebAssembly) · ROS 2 Humble C++ nodes · Quantization / pruning / distillation · Latency and memory benchmarking · Docker
MLOps & Systems	DVC pipelines · MLflow · Weights & Biases · GitHub Actions CI · pytest · ruff · Signed OTA bundles · MQTT canary rollout with automatic rollback · Kafka · Spring Boot / Java 21 · FastAPI · Postgres/pgvector · React · Docker Compose · Distributed GPU training (Slurm, FSDP)
LLM / VLM	Qwen2.5-VL LoRA fine-tuning · Vision-language action grounding with deterministic safety overrides · On-premise RAG (Ollama, hybrid dense + lexical retrieval, citation and refusal evaluation) · Agentic tool use over MCP · vLLM serving
Safety & Process	ISO 26262 (ASIL-B) · ISO 21448 (SOTIF) · MISRA C++ · Hazard analysis and release gates · Automated KPI pipelines · Simulation-based validation · HIL tooling
Languages	Python · C++ · Rust · C · SQL · TypeScript · English (C1) · German (B2) · Kannada · Hindi

Generated from the content of pavanyadava007.github.io on 2026-09-23. Benchmarks were measured on an NVIDIA L4 and an AMD EPYC 7R13 on an AWS EC2 host; nothing is extrapolated to hardware not run on, and each project's write-up lists what failed or could not be measured.